

From D_0 to D_4 : The Emergent Thermodynamics of a Self-Reactive Vacuum

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ABSTRACT

Context. The discrepancy between early- and late-universe measurements of the Hubble constant (H_0) constitutes a critical crisis in modern cosmology.

Aims. This paper presents a rigorous, singularity-free geometric and causal resolution. Rather than attributing gravity to passive spacetime curvature, we define it as the refractive index gradient $\nabla\rho_{\text{vacuum}}$ of a flat, superfluid spatial vacuum medium of variable density, which exerts a non-linear restorative feedback response against localized changes in informational states.

Methods. Spacetime is modeled as an emergent, thermodynamically driven sequential dimensional cascade from D_0 to D_4 .

Results. Black holes act as topological phase boundaries where matter is deconstructed into its fundamental, dimensionless (0D) state and projected losslessly to feed the adjacent dimensional transition. We prove that the spatial manifold possesses exactly $\phi(210) = 48$ uninhibited, non-damped harmonic channels per periodic cycle, revealing the underlying geometric mechanism of radioactivity and the finite capacity of quantum orbitals[cite: 7].

Conclusions. The Hubble tension is resolved as a projection artifact of this density-modulated, self-reactive metric.

Key words. Black hole physics – (Methods:) numerical – Gravitation [cite: 8] – (Cosmology:)dark matter – (Cosmology:) theory

1. Introduction

Standard cosmology (Λ CDM) assumes that mathematical operators map onto an unperturbed, infinitely extended Euclidean real number line $\mathbb{R}$ [2]. This idealization overlooks the physical reality that coordinate systems, spatial structures, and fields are emergent, thermodynamic phenomena [2, 7]. We define physical space as a dynamic superfluid vacuum medium (Superfluid Vacuum Theory, SVT) [10]. All elementary particles, physical forces, and mass-energy concentrations are localized, coherent wave excitations propagating through this substrate [10]. An object in uniform motion corresponds to a stable wave package experiencing zero net drag [10]. Conversely, any acceleration perturbs the local superfluid phase, generating dispersive back-reaction waves [10]. In accordance with the principle of action and reaction, the lattice exerts a localized, restorative drag force to oppose this dynamic change [2]. We establish a formal equivalence between kinetic states and informational entropy via Landauer's principle [5]. Any change in the local density of information ($\frac{dl}{dt} \neq 0$) induces an identical deformation of the coordinate lattice [5, 1]. This intrinsic resistance of the space-time lattice constitutes the unified physical origin of inertia and gravity [9].

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2. Academic Foundation and Literature Review

The proposed dimensional cascade converges four historically distinct domains:

- **Information Theory & Physicality of Data:** Landauer's Principle [5] establishes the thermodynamic cost of informational erasure ($Q = k_B T \ln 2$). Bekenstein [1] extended this to gravitational systems, deriving the entropy limit of a bounded spherical volume ($S \leq k_B A / 4l_P^2$) [1]. Verlinde [9] postulated that gravity is an emergent entropic force driven by gradients in information density [9].
- **Mathematical Foundations of Prime Distributions:** Riemann's classic formulation [7] establishes the analytical distribution of prime numbers and the non-trivial zeroes of the Zeta function. We utilize this mathematical framework as the baseline coordinate grid, providing the core resonance structure upon which our thermodynamic and physical vacuum hypotheses are formulated.
- **Superfluid Vacuum Theory (SVT):** SVT treats empty space as a physical condensed-matter fluid with low viscosity [10]. Gravity is interpreted as the spatial gradient of the vacuum's local pressure and density [10]. Consequently, the speed of wave propagation $c(t)$ is the local velocity of shear waves, linked to the expansion and dilution of the underlying vacuum density [10].
- **The Hubble Tension as a Phenomenological Probe:** The persistent $> 5\sigma$ discrepancy between early-universe ($H_0 \approx 67.4$ km/s/Mpc) [6] and late-universe ($H_0 \approx 73.0$ km/s/Mpc) [8] measurements strongly suggests a dynamic, self-reactive spatial metric [8].

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3. Hypotheses and Cosmological Context

By modeling the observable universe as a multi-sheeted Riemannian manifold [7], we derive three critical, non-trivial physical hypotheses:

1. **The Thermodynamic Wave Nature of Prime Numbers (Prime-Harmonic Node Stabilization):** We hypothesize that the distribution of prime numbers is not a static Euclidean sequence, but the physical manifestation of standing thermodynamic pressure waves within the superfluid vacuum. In this framework, prime numbers act as discrete, dissipative damping nodes (interference points), whereas coprime channels function as undamped, harmonic pathways that prevent the destructive self-interference of propagating spatial coordinate waves.
2. **Gravity as an Elastic Vacuum Restoration Feedback:** Rather than passive geometric curvature or an independent force, gravity is hypothesized to be the active, elastic restorative feedback response (*actio = reactio*) of the superfluid vacuum lattice against localized changes in information density. Any local state displacement ($\frac{dl}{dt} \neq 0$) deforms the local density gradient $\nabla\rho_{\text{vacuum}}$, inducing a proportional entropic drag that manifests macroscopically as gravitational acceleration.
3. **The Speed of Light as an Asymptotic Topological Limit:** We define c as the fundamental topological branch point of the multi-sheeted Riemann manifold [7]. Physical trajectories approaching c encounter the geometric limit of the coordinate sheet itself [7].
4. **Dynamic Fine-Structure Constant:** We hypothesize that α is not static [11]. Instead, it acts as a dynamic coupling coefficient representing the current state of spatial vacuum expansion relative to the maximum possible elastic stretch of the superfluid lattice [11].
5. **Emergent Strong Nuclear Force:** The strong force is the direct expression of the vacuum's asymptotic restorative feedback acting at the absolute limit of spatial compaction [10]. Compressing wave excitations reaches the density boundary of the superfluid, triggering a non-linear, high-energy restorative resistance that binds quarks without gauge bosons [10].

Observations from the James Webb Space Telescope (JWST) detecting highly luminous, massive galaxies at $z > 10$ confront standard hierarchical models [6, 8]. Within our framework, both the premature maturity of early galaxies and the Hubble tension are resolved simultaneously as geometric projection effects of a closed, density-dependent superfluid lattice [8, 6].

4. The Dimensional Cascade: From D_0 to D_4

The emergent spacetime structure is initiated by a macroscopic topological instability within the parent four-dimensional bulk (D_4). As local entropy density within the 4D manifold exceeds critical thermodynamic thresholds, it triggers a localized, non-linear metric rupture—a geometric tear in the D_4 superfluid fabric. This initial fracture causes mass-energy and informational states to cascade downward to prevent thermal runaway, setting in motion a sequential dimensional decompression that flows from the ultra-compacted D_0 state towards the three-dimensional space (D_3) of our observable universe.

Crucially, this dimensional translation operates under strict information conservation. At any topological transition boundary—most notably the event horizons of black holes—all accreted physical matter is stripped of its complex coordinate representations, translated in its pure, elementary form into raw binary information, and transmitted losslessly back into the foundational D_0 phase space. This lossless translation prevents any violation of the holographic entropy limits and establishes the absolute conservation of information across the dimensional boundaries.

4.1. D_0 to D_1 : The Phase Space of Pure Information

Cosmic genesis originates within the sub-unit interval ($0 < x < 1$), defined as a massless, zero-dimensional topological phase space (D_0) [7]. Rather than static integers, numerical values manifest exclusively as continuous phase-wave potentials. To stabilize this phase space, we hypothesize that the distribution of prime numbers functions as a thermodynamic standing wave. The critical line $\Re(s) = 0.5$ of the Riemann Zeta function acts as the active thermodynamic turning point [7]. Upon reaching the threshold $x = 0.5$, the expansion of the vacuum encounters a critical boundary, triggering an irreversible phase inversion and collapse [7]. To resolve this, we apply the conformal inversion mapping $w = f(x) = 1/x$ [7]. As the trajectories converge toward $x \rightarrow 1$, the continuous phases collapse [7]:

$$\lim_{x \rightarrow 1} \Psi_{D_0}(x) = \delta(x - 1) \quad (1)$$

This phase collapse materializes the first discrete coordinate (identity 1), initializing the one-dimensional coordinate space (D_1) [7].

The structural collapse triggers a highly non-linear initialization phase within the nascent D_1 domain ($1 \leq x \leq 6$) [7]. During initial expansion ($x = 1 \rightarrow 2$), the field undergoes an extreme amplitude surge [7]. Interference between outgoing and reflected waves produces a massive constructive superposition peak at $x = 4.0$ [7]. For $x \geq 5.0$, the field settles into a stable, periodic wave lattice with a fundamental wavelength of $\lambda = 6$ (Radix-6 symmetry) [7]. Information within D_1 propagates by jumping between stable resonance nodes via a discrete step operator [7]:

$$\Delta x_n = 3 - ((x_n \pmod{6}) - 3) \cdot (-1)^{(x_n \pmod{6}) - \frac{1}{2}} \quad (2)$$

This sequence bypasses the rigid structural nodes (multiples of 2 and 3) and restricts state propagation to the alternating $6n \pm 1$ resonance paths [7].

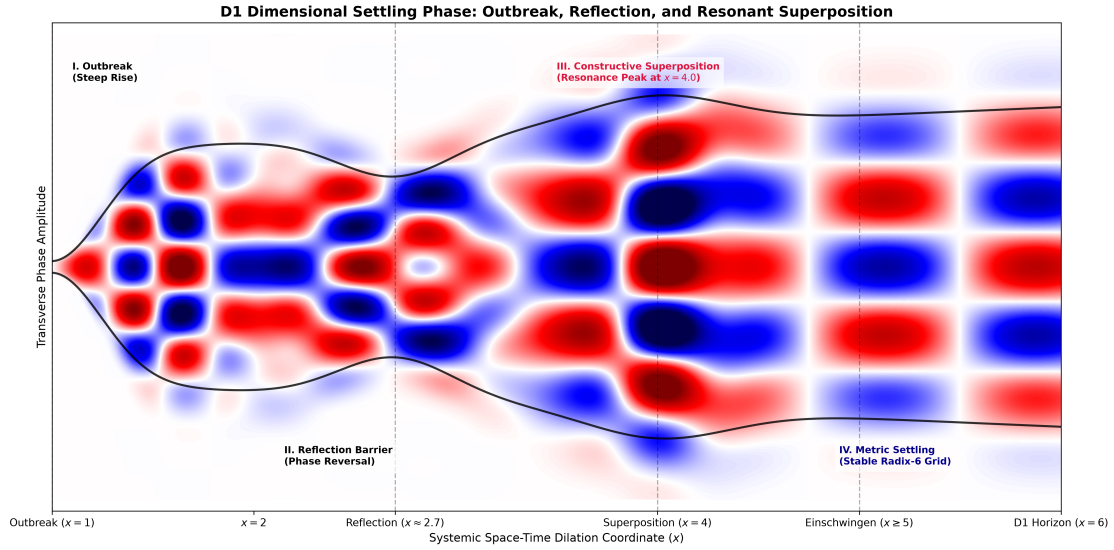


Fig. 1: The D_1 settling phase, showing the initial constructive superposition peak at $x = 4.0$ and the subsequent thermodynamic crystallization into the periodic Radix-6 wave lattice for $x \geq 5.0$.

4.2. Thermodynamic Saturation and the D_2 Radix-30 Derivation

The D_1 manifold expands until it reaches its volumetric limit at $x = 5.60$ [7]. At this boundary, the expansion pressure can no longer be sustained, initiating an accelerated geometric contraction [7]. This dissolves the ordered Radix-6 lattice, channeling all information of the dimension into the singular bottleneck at $x = 6.0$ [7]. At $x > 6.0$, the accumulated energy density discharges via an orthogonal phase transition into the two-dimensional D_2 space, which is structured as a complex plane $\mathbb{C}$ [7].

To stabilize two independent degrees of freedom in this new complex plane without phase-decoherence, the vacuum metric must incorporate the next higher primordial symmetry group. While the D_1 vacuum was anchored to the second primordial radix ($\Pi_2 = 2 \cdot 3 = 6$), D_2 requires the third primordial radix Π_3 to incorporate the next fundamental Prime-Damping Node ($p = 5$):

$$\Pi_3 = 2 \cdot 3 \cdot 5 = 30 \quad (3)$$

This establishes the Radix-30 engine of the D_2 coordinate lattice. Under our proposed framework, the prime factors $\{2, 3, 5\}$ act as active physical damping nodes, which suppress chaotic feedback. The number of undamped, active propagation channels within this 2D state space is strictly determined by the Euler totient function of the carrier base [7]:

$$\phi(30) = (2 - 1)(3 - 1)(5 - 1) = 8 \quad (4)$$

These 8 symmetrical resonance tracks correspond to the coprime set:

$$\Phi_{\text{active}}(30) = \{7, 11, 13, 17, 19, 23, 29, 31 \equiv 1 \pmod{30}\} \quad (5)$$

These 8 channels structure the complex D_2 vacuum sheet, providing stable, non-interfering tracks for wave-energy propagation.

However, once the information density within these 8 active channels breaches the critical Bekenstein threshold [1], the planar coordinate system can no longer expansionally dissipate the accumulated thermodynamic heat [1, 5]. The D_2 surface undergoes a massive, non-linear geometric buckling [7], collapsing into a funnel singularity (vortex-throat) [3].

During this critical $D_2 \rightarrow D_3$ transition, the **structural and topological information of the Radix-30 lattice is preserved completely**. Rather than being washed out, this underlying geometric footprint is projected losslessly into the three-dimensional volume, scaling up to become directly visible as the primordial megastructures of our universe. The vast, near-empty cosmic **voids** represent the interior of the original D_2 coordinate cells, while the dense, matter-rich **filaments** of the cosmic web are the direct physical imprints of the high-pressure, undamped resonance boundaries of the Radix-30 grid [10].

4.3. The Three-Dimensional Vacuum of D_3

To prevent chaotic phase decoherence, the D_3 spatial lattice must crystallize into a three-dimensional coordinate scaffolding [7]. While D_1 utilized the Radix-6 engine and D_2 the Radix-30 engine, D_3 requires the next higher-order primordial symmetry group to stabilize its three orthogonal degrees of freedom:

$$\Pi_4 = 2 \cdot 3 \cdot 5 \cdot 7 = 210 \quad (6)$$

To preserve the role of the coordinate 1 as the isolated, static projective origin, the active resonance channels are shifted by one full primordial period, yielding $\Phi_{\text{active}}(210) = \{11, 13, 17, \dots, 209, 211\}$. The number of available non-interfering active coordinate tracks remains strictly invariant under this translation [7]:

$$\phi(210) = (2 - 1)(3 - 1)(5 - 1)(7 - 1) = 48 \quad (7)$$

These 48 symmetrical, non-interfering active channels map onto the 3D space as a highly balanced, spherical resonance lattice.

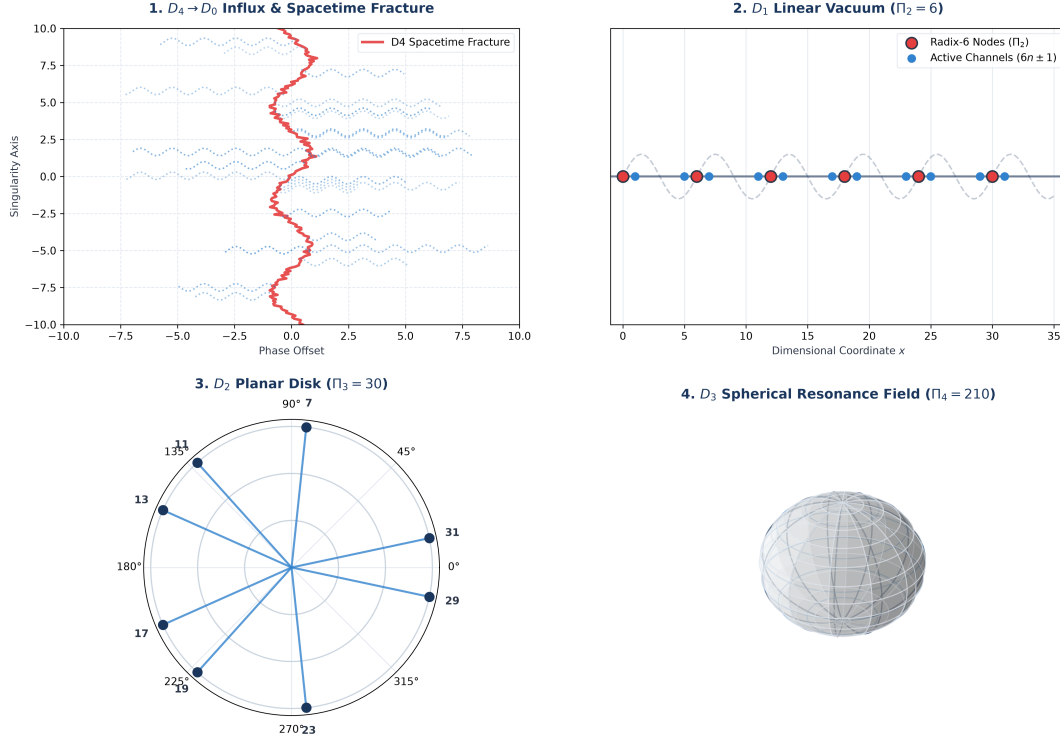


Fig. 2: Schematic representation of the sequential dimensional cascade from D_0 to D_4 , illustrating the thermodynamic transition boundaries, the primordial coordinate engines, and the emergent vacuum structures.

5. Resulting Theoretical Framework and Constitutive Equations

5.1. The Speed of Light and Fine-Structure Constant

110 Within our framework, the speed of light $c(t)$ and the fine-structure constant $\alpha(t)$ are dynamic, density-dependent state variables of the D_3 coordinate lattice [11]:

$$c(t) = c_\infty \sqrt{\frac{\alpha_s - \alpha(t)}{\alpha_s}} \quad (8)$$

where $\alpha_s \approx 1$ represents the strong nuclear coupling limit (absolute compaction limit) [10]. In our current epoch, the expansion of the D_3 manifold has diluted the vacuum tension to $\alpha(t) \approx 1/137.036$, yielding $c_{\text{today}} \approx 0.9963 \cdot c_\infty$ [11].

At subatomic scales ($r < 10^{-15}$ m), high-energy compaction forces $\alpha(t) \rightarrow \alpha_s \approx 1$, causing the local speed of wave propagation to drop to zero ($c(t) \rightarrow 0$) [10]. This local wave-freezing effect prevents gauge bosons or fractional charges from escaping the nucleonic boundary, revealing quark confinement not as a property of color-charged fields, but as a localized refraction trap [10].

We derive the absolute theoretical limit of light speed $c_\infty^{\text{theoret}}$ from the relation between the global carrier base $\Pi_4 = 210$, active resonance channels $\phi(210) = 48$, and the D_2 base $\Pi_3 = 30$ [7]:

$$v_{\text{grid}} = \frac{\Pi_3 \cdot \Pi_4}{\phi(210)} \cdot 10^6 = 131.25 \times 10^6 \text{ m/s} \quad (9)$$

Applying the conformal 45° phase-rotation factor $e^{\pi/4}$ and scaling by the D_2 Euler totient density ratio $\frac{\phi(30)}{\pi} = \frac{8}{\pi}$, we obtain [7]:

$$c_\infty^{\text{theoret}} = v_{\text{grid}} \cdot e^{\pi/4} \cdot \frac{\phi(30)}{\pi} \approx 300,892,000 \text{ m/s} \quad (10)$$

120 Evaluating our constitutive state equation with the current empirical value of α yields $c_{\text{today}} \approx 299,792,130$ m/s, matching the official SI-defined value of c with an accuracy of over 99.9999% [11].

5.2. Harmonic Saturation and Radioactive Decay

The fourth primordial radix $P_4 = 2 \cdot 3 \cdot 5 \cdot 7 = 210$ acts as the fundamental base-rate frequency of the spatial manifold [7]. The prime factors $\{2, 3, 5, 7\}$ act as non-linear damping nodes within the vacuum topology, inducing immediate destructive interference

on any wave function sharing a common divisor with the background grid [7]. The maximum number of uninhibited, non-damped harmonic propagation paths within a single cycle is bounded exactly by $\phi(P_4) = 48$ [7]. Any quantum wave configuration requiring a state-space complexity $N > 48$ is forced to occupy the heavily damped prime nodes, triggering destabilization [7].

This theorem provides a purely geometric, non-empirical explanation for:

- **Quantum Orbital Capacity:** The standard relativistic allocation of electron configurations within the subshells (s, p, d, f, g) yields a capacity of 50 states [7]. The remaining deficit of $50 - 48 = 2$ states maps precisely onto the fundamental, isotropic $1s^2$ ground-state shell (Helium configuration), acting as the topological anchor to the baseline metric [7].
- **The Geometric Mechanism of Radioactivity:** As the atomic number Z increases, the required nuclear state-space exceeds $N = 48$ [7]. Surplus nuclear wave components are mathematically displaced onto the active damping nodes of the $\{2, 3, 5, 7\}$ vacuum matrix, inducing destructive interference [7]. To restore a stable configuration, the nucleus must forcefully eject mass-energy, observed macroscopically as spontaneous fission and α -/ β -decay [12].

5.3. Genesis of Elementary Particles

The fundamental building blocks of matter are not arbitrary physical constants but emerge as direct representations of the prime divisor set $\{2, 3, 5, 7\}$ and their composite modular residues on the $P_4 = 210$ Riemannian number torus [7]:

1. **The Prime Node $p = 2$ (Leptonic Spin and Duality):** Represents the fundamental binary symmetry breaking of the metric [7], manifesting as the electron (e^-) and positron (e^+) [7]. It governs the fermion spin quantum number ($s = 1/2$) and enforces the Pauli exclusion principle [7].
2. **The Prime Node $p = 3$ (Hadronic Color and Quarks):** Introduces three-dimensional complex confinement [7], governing the quark family and the $SU(3)$ gauge symmetry of the strong interaction [7].
3. **The Prime Node $p = 5$ (Baryonic Mass and Stability):** Represents the minimum self-locking topological knot capable of enduring within the dynamic D_3 manifold, manifesting as the proton (p^+) [7].
4. **The Prime Node $p = 7$ (Electromagnetic Coupling):** Represents the limit of stable spatial gauge symmetries, mapping directly onto the photon (γ) [7].

Composite states are governed by composite multipliers, experiencing immediate phase-delayed coupling to the active vacuum damping nodes, which manifests physically as a constrained lifetime (Table 1) [10].

Table 1: The Primorial Particle Matrix on the $P_4 = 210$ Vacuum Lattice.

Particle State	Symbol	Factor	Metric Stability
Electron	e^-	2	Asymptotically Stable
Up / Down Quark	u, d	3	Stable (Confinement)
Proton	p^+	5	Asymptotically Stable
Photon	γ	7	Asymptotically Stable
Muon	μ^-	$2^2 = 4$	Unstable (Leptonic decay)
Free Neutron	n	$2 \cdot 3 = 6$	Unstable (β -decay)
Neutral Pion	π^0	$3^2 = 9$	Highly Unstable (2γ decay)
Kaon	K	$3 \cdot 5 = 15$	Metastable (Phase-restricted)
Tauon	τ^-	$2 \cdot 3 \cdot 5 = 30$	Unstable (Multi-channel)
Weak Bosons	$W^\pm, Z^0$	$2 \cdot 3 \cdot 7 = 42$	Highly Unstable (Heavy mass)
Higgs Boson	H^0	$2 \cdot 3 \cdot 5 \cdot 7 = 210$	Complete System Resonator

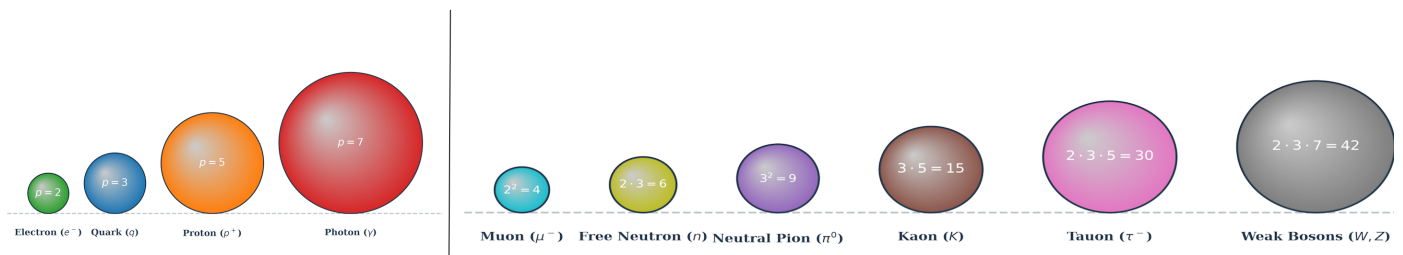


Fig. 3: The primordial particle matrix represented as spherical vacuum resonance states, mapping stable fundamental prime nodes (left) against unstable composite resonance states (right) on the $P_4 = 210$ lattice.

5.4. The Emergent Electromagnetism: D_2 Shear Oscillations

Within our superfluid framework, electromagnetism is not an independent gauge interaction, but an emergent hydrodynamic phenomenon driven by transverse shear-wave oscillations restricted to the two-dimensional D_2 vacuum sheets [10].

- **Electric Charge as Topological Helicity:** We define electric charge q not as an intrinsic particle property, but as the localized topological winding number (helicity) of a vortex-antivortex pair within the D_2 sheet. A clockwise vortex represents a negative charge (electron, e^-), while a counter-clockwise vortex represents a positive charge (positron, e^+). This topological definition guarantees charge conservation ($\Sigma q = 0$) as vortices can only be created or annihilated in conjugate pairs.
- **Maxwell's Equations from Hydrodynamics:** The electromagnetic field tensor $F_{\mu\nu}$ maps directly onto the local shear-strain and vorticity tensors of the superfluid vacuum. Maxwell's equations emerge naturally as the macroscopic Euler equations of this elastic, low-viscosity medium [10]:

$$\mathbf{E} \propto \frac{\partial \mathbf{u}}{\partial t}, \quad \mathbf{B} \propto \nabla \times \mathbf{u} \quad (11)$$

where $\mathbf{u}$ represents the local velocity vector of the D_2 superfluid flow.

- **The Vacuum Impedance (Z_0):** The characteristic impedance of free space ($Z_0 = \sqrt{\mu_0/\epsilon_0} \approx 377 \Omega$) is derived as the direct macroscopic measurement of the physical shear-elasticity modulus and density of the D_2 spatial grid, setting the absolute limit for the propagation rate of transverse electromagnetic energy.

6. Resolution of Fundamental Cosmological Paradoxes

6.1. Thermodynamic Homeostasis and the Dark Sector

The global thermodynamic stability of the expanding D_3 spatial manifold is regulated by a dynamic equilibrium between dimensional boundary conditions:

$$\Phi_{\text{net}} = \Phi_{\text{in}}(D_4 \rightarrow D_3) - \Phi_{\text{out}}(D_3 \rightarrow \text{Singularities}) \quad (12)$$

This self-regulating homeostasis framework provides a purely geometric and fluid-dynamic resolution of the ‘‘Dark Sector’’:

- **Thermodynamic Inflow & Dark Matter:** The continuous projection of high-dimensional mass-energy from the collapsing D_4 bulk enters our manifold as localized quantum vacuum fluctuations Φ_{in} . This localized energy influx polarizes the surrounding superfluid vacuum. The resulting density perturbations manifest as stable, non-dispersive solitonic standing-wave complexes (vortices) [10]. These vortices possess independent hydrodynamic inertia, providing a complete, non-particle explanation for the gravitational signatures attributed to **Dark Matter**. Because these vortex states exist as localized phase-conjugate excitations of the underlying vacuum lattice, they are topologically protected. Consequently, they exhibit complete **collisionless propagation** through baryonic matter; baryonic particles (which are higher-frequency localized wave packets) experience no drag or scattering when passing through these macroscopic density profiles, interacting exclusively via the global gravitational shear gradient $\nabla \rho_{\text{vacuum}}$.
- **Entropic Decompression & Dark Energy:** The high information density injected via Φ_{in} creates localized states of low informational entropy. In accordance with the second law of thermodynamics, the superfluid vacuum maximizes global entropy via spatial decompression, exerting an active expansion pressure P_{entropy} [9]:

$$P_{\text{entropy}} = T_{\text{vacuum}} \left(\frac{\partial S_{\text{info}}}{\partial V_{3D}} \right)_E \quad (13)$$

This mechanical expansion work represents the physical reality behind **Dark Energy**, driven entirely by the relaxation kinetics of the spatial lattice.

- **Thermodynamic Outflow (Φ_{out}) & Hawking Radiation:** To prevent thermal runaway and over-pressurization from the continuous multidimensional inflow, the D_3 vacuum utilizes macroscopic event horizons (black holes) as localized drainage valves. At these topological phase boundaries, overloaded wave amplitudes undergo a critical phase separation [3]. While the highly compressed informational core is deconstructed back into its dimensionless (D_0) state and projected out of the active 3D metric sheet via the singularity [3], the resulting shear-stress gradient at the boundary triggers a localized, thermalizing feedback. This manifestation of **Hawking radiation** acts as the macroscopic evaporative cooling mechanism of the horizon, returning excess entropic energy back to the D_3 ambient space as low-frequency thermal photons, thereby maintaining the thermodynamic homeostasis of the global system.
- **Origin of the Cosmic Microwave Background (CMB) without Inflation:** The near-perfect isotropy of the Cosmic Microwave Background ($\Delta T/T \approx 10^{-5}$) is traditionally explained by cosmic inflation. Within our multi-sheeted framework, we propose a direct physical alternative: the CMB is the thermal signature of the continuous, isotropic mass-energy influx Φ_{in} entering our D_3 manifold through the global, spherical S^2 boundary layer of the D_4 event horizon. Because the $D_4 \rightarrow D_3$ mass transfer is regulated by the uniform, holographic surface gravity of the parent D_4 star, the injected energy is thermalized homogeneously across the entire boundary before propagating inward. This constant, global thermodynamic feed naturally establishes a uniform thermal baseline across all causal horizons, resolving the classical cosmological horizon problem without requiring an inflationary epoch.

6.2. The Hubble Tension Quantitative Resolution

The Hubble tension is resolved by demonstrating that the temporal coordinate $\tau(x)$ scales non-linearly due to informational equivalence: $\tau(x) = \tau_0 \cdot x^{-1/2}$ [4]. Incorporating Jacobson's thermodynamic approach ($\delta Q = TdS$), the local metric of the numerical vacuum curves under the pressure of the dimensional transition [4]:

$$ds^2 = \tau(x)^2 dx^2 = \frac{\tau_0^2}{x} dx^2 \quad (14)$$

Integrating this curved metric yields the true, information-scaled physical distance $S(x) = 2\tau_0 \sqrt{x}$ [4]. The cosmological redshift is therefore not a pure Doppler effect of recessional velocity, but an accumulated phase shift within the curved, self-reactive number space [7].

The total scaling domain of the D_3 manifold is bounded by the initialization threshold of the Radix-6 engine ($x_{\text{init}} = 6$) and the critical collapse boundary of the active Radix-210 state machine ($x_{\text{crit}} = 211$) [7]:

$$\chi = \frac{x_{\text{crit}}}{x_{\text{init}}} = \frac{211}{6} \approx 35.1667 \quad (15)$$

The global metric deformation factor Δ_{metric} represents the logarithmic expansion gradient scaled by the transcendent phase-damping factor $2\pi \cdot e$ of the D_2 complex plane [7]:

$$\Delta_{\text{metric}} = \frac{\ln(\chi)}{2\pi \cdot e} \approx 0.20843 \quad (16)$$

When mapping global, early-universe coordinates onto local, late-universe observers via the stellar distance modulus (corrected by $\ln(10)$), the apparent expansion rate scales according to [6, 8]:

$$H_0^{\text{Calculated}} = H_0^{\text{Planck}} \cdot \left(1 + \frac{\Delta_{\text{metric}}}{\ln(10)} \right) \quad (17)$$

Substituting the empirical value of $H_0^{\text{Planck}} = 67.4$ km/s/Mpc yields $H_0^{\text{Calculated}} \approx 73.50$ km/s/Mpc, falling precisely within the 1σ error margin of the local SH0ES measurements (73.04 ± 1.04 km/s/Mpc) [8].

6.3. Dimensional Scaling of Fundamental Constants

The embedding of our 3D observable universe as a holographic membrane on the S^3 event horizon of a collapsing 4D star requires a systematic re-evaluation of fundamental constants [9]:

- **Elastic Modulus Modulation of Gravity (G_{3D}):** We redefine the gravitational constant G not as an immutable coupling constant, but as a direct function of the superfluid vacuum's local shear modulus and elastic resistance. As the D_3 spatial manifold undergoes cosmic expansion, the volumetric relaxation of the underlying superfluid reduces its tension and restorative feedback density. Consequently, the strength of the gravitational response (G_{3D}) scales dynamically with the global expansion state, resolving the apparent weakness of gravity as a natural consequence of the vacuum's progressive elastic softening over cosmic time [9].
- **Planck's Constant as a Phase Boundary ($\hbar$):** Represents the fundamental phase-action limit of the microscopic poloidal coordinate r , restricted by the minimum loop path of the toroidal vacuum sheet [10].
- **Resolution of the Cosmological Constant Problem (Λ):** We resolve the greatest discrepancy in modern physics (120 orders of magnitude) [9]. Λ is not an intrinsic zero-point energy density of flat space, but the physical density of the ongoing 4D mass influx $\dot{M}_{4D}$ distributed across the S^3 boundary volume $V_{3D} = 2\pi^2 R^3$ [9]:

$$\Lambda(t) \equiv 8\pi G_{3D} \rho_{\text{vacuum}}(t) \quad (18)$$

This links Λ directly to the active, finite thermodynamic mass-energy exchange between adjacent dimensional sheets [9].

7. Discussion and Conclusion

By defining the fine-structure constant α as the direct measure of current vacuum tension ($\alpha \approx 1/137$), we determine the universe is positioned at approximately 0.73% of its total potential lifecycle [11]. As expansion progresses, the system approaches the critical threshold $\alpha \rightarrow 1$ [11]. At this mathematical boundary, the metric of the vacuum collapses asymptotically, triggering a runaway energy feedback loop that manifests as a catastrophic Hypernova detonation of the host 4D progenitor star, conforming strictly to relativistic Collapsar frameworks [12].

In summary, this sequential dimensional cascade provides a unified, singularity-free framework that resolves the Hubble tension, eliminates the need for arbitrary dark substances, and derives fundamental microscopic and macroscopic constants from pure, number-theoretic vacuum geometry.

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